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REVIEW

Natural Language Processing of Psychotherapy Sessions

A narrative review of automated fidelity coding, empathy detection, and feedback to therapists

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ABSTRACT

Process research in psychotherapy has long depended on human coding of session recordings, a reliable but labor-intensive procedure that rarely reaches routine care. Natural language processing (NLP) offers the prospect of analyzing therapeutic conversations at scale. This narrative review summarizes the main lines of research on the automated analysis of psychotherapy sessions. We first describe the automated coding of motivational interviewing fidelity, from early text classification studies to automated pipelines that combine speech recognition, speaker identification, and behavioral coding. We then review work on empathy detection and on other process variables such as the working alliance, and describe systems that return automatically generated metrics to therapists or offer suggestions during text-based conversations. The evidence suggests that selected therapist behaviors can be coded with promising agreement with human raters, whereas relational constructs remain more difficult to capture. We discuss limitations regarding validity, generalizability across settings and languages, possible reactivity to metrics, and the protection of highly sensitive session data. From a meaning-centered perspective, we argue that such tools can support reflective practice but must not replace clinical judgment or the personal encounter at the heart of psychotherapy.

Keywords: natural language processing · motivational interviewing · treatment fidelity · empathy · psychotherapy process research

1 Introduction

What happens in the therapy room has long been difficult to study at scale. Process research depends on recordings that trained raters code according to behavioral manuals, a procedure that is reliable when done well but time-consuming and costly. As a consequence, routine care rarely receives systematic feedback on the quality of therapeutic conversations, and process research remains restricted to comparatively small samples (Atkins et al., 2014).

Natural language processing (NLP), the computational analysis of spoken and written language, promises to change this situation. Over the past decade, research groups have developed models that transcribe sessions, identify speakers, and assign behavioral codes to individual utterances. This narrative review summarizes the main lines of this research, focusing on motivational interviewing (MI) fidelity, empathy detection, and feedback to therapists, and then discusses questions of validity and privacy that must be answered before such tools enter routine practice. The review is selective rather than systematic; a broader mapping of machine learning in psychotherapy research is provided by Aafjes-van Doorn et al. (2021).

2 Why Process Coding Matters

Treatment fidelity, meaning the extent to which a therapist delivers an intervention as intended and with skill, is central to both research and training. Meta-analytic evidence on the association between therapist adherence or competence and outcome is less consistent than often assumed (Webb et al., 2010), yet fidelity monitoring remains indispensable for interpreting trials and for ensuring that what is disseminated corresponds to what was tested. Therapist empathy, in turn, is among the relationship

factors most consistently associated with outcome (Elliott et al., 2018).

MI offers a particularly suitable test case for automated coding. It is a well-specified, conversation-based approach in which the therapist's verbal behavior, such as open questions, complex reflections, and affirmations, is thought to shape the client's language about change (Miller & Rollnick, 2013). Its fidelity is assessed with established coding systems such as the Motivational Interviewing Treatment Integrity code, whose current version combines global ratings with counts of specific behaviors (Moyers et al., 2016). Human coding with such instruments requires extensive training and substantial time per session, which limits its use in routine care.

3 Automated Coding of Motivational Interviewing Fidelity

Atkins et al. (2014) provided an early demonstration that statistical text classification can approximate human MI codes. Using transcripts of counseling sessions, they trained topic models to predict utterance-level codes and session-level ratings and reported that the models reproduced human coding reasonably well for several, though not all, codes. Tanana et al. (2016) subsequently compared different NLP methods for coding MI utterances and found that agreement with human coders varied considerably across codes, with some behaviors recognized more reliably than others.

Later work moved from manually produced transcripts to automated pipelines. Flemotomos et al. (2022) described a system that processes audio recordings through voice activity detection, speaker diarization, automatic speech recognition, and role identification before assigning behavioral codes and estimating session-level quality indicators. Applied to recordings from a real-world counseling setting, such a pipeline makes it possible, in principle, to evaluate large numbers of sessions without human transcription. Multistage pipelines also raise a specific problem: errors can propagate across stages, so that inaccuracies in speech recognition or speaker attribution may affect the resulting codes.

Taken together, these studies suggest that automated MI coding is feasible and that agreement with human raters can approach human inter-rater reliability for some codes and session-level indicators. At the same time, performance is uneven, and most models were developed and tested within a limited number of datasets, settings, and languages, predominantly English. Whether models trained in one context generalize to other populations, treatment approaches, and languages remains largely open.

4 Empathy Detection and Other Process Variables

Empathy is harder to operationalize than specific MI behaviors, because it depends on the meaning of an utterance in context rather than on surface features alone. Xiao et al. (2015) nevertheless showed that observer-rated therapist empathy in addiction counseling sessions could be predicted from speech and language features to a degree that suggested practical potential. For text-based peer support, Sharma et al. (2020) proposed a framework that distinguishes emotional reactions, interpretations, and explorations as communication mechanisms of empathy, and developed models to identify them in online conversations.

Beyond empathy, NLP methods have been applied to other process constructs. Goldberg et al. (2020) used session transcripts to predict client-rated working alliance and found that the models captured part of the variance in alliance ratings, although predictive accuracy was modest. The scoping review by Aafjes-van Doorn et al. (2021) documented a rapidly growing but methodologically heterogeneous field, with frequent use of small samples and limited external validation.

These findings illustrate a basic tension. Constructs such as empathy and alliance are relational and partly subjective; they are experienced by clients and inferred by observers. A model that predicts observer ratings from text learns the regularities of a particular coding tradition, not empathy as such. Its outputs should therefore be interpreted as estimates of how trained raters would likely judge a conversation, with all the assumptions that this entails.

5 From Measurement to Feedback

The main practical promise of automated coding lies in feedback. Imel et al. (2019) developed a prototype that provided counselors with automatically generated MI fidelity metrics after a session and examined its design feasibility; participants generally regarded the feedback as understandable and potentially useful for professional development. Such systems could make performance-based feedback, which is otherwise available mainly in supervised training, accessible on a continuous basis.

A further step is real-time support. Sharma et al. (2023) reported, in a randomized trial on an online peer support platform, that an AI system suggesting how to make responses more empathic increased the

rated empathy of conversations, particularly among supporters who had difficulty expressing empathy. This study concerned peer supporters rather than psychotherapists, and empathy was assessed at the level of written messages, so its implications for clinical practice should not be overstated. It nevertheless illustrates that the line between measurement and intervention becomes blurred once models offer suggestions during a conversation.

6 Validity, Privacy, and Ethical Questions

Several questions of validity must be addressed before automated coding can be used for consequential decisions. First, agreement with human coders is a necessary but insufficient criterion; human codes themselves have limited reliability, and a model that reproduces them also reproduces their errors and biases. Second, predictive validity, that is, whether automated indicators relate to client outcomes, has rarely been examined. Third, performance must be tested across client groups, dialects, and languages, since speech recognition and language models may work less well for speakers who are underrepresented in training data. Finally, if fidelity scores were used for evaluation or reimbursement, therapists might adapt their verbal behavior to the metric rather than to the client.

Privacy is no less important. Therapy recordings contain highly sensitive health information about clients and, frequently, about third parties. In the European Union, such data fall under the special categories of personal data protected by the General Data Protection Regulation, which requires a valid legal basis, purpose limitation, data minimization, and appropriate technical and organizational safeguards (Regulation (EU) 2016/679, 2016). Recording sessions for automated analysis requires explicit and informed consent, secure storage, and clear rules on retention and secondary use, for example for model training. Local processing and de-identification can reduce, but not eliminate, these risks, since the content of a conversation may itself be identifying.

Finally, the advent of large language models changes the field. Such models can generate summaries, feedback, and codes with little task-specific training, but they also produce fluent text whose relation to the actual conversation is not guaranteed (Bender et al., 2021). Validation standards established for specialized coding models, including comparison with human ratings and transparency about error rates, should apply equally to these more general systems.

7 Conclusion

NLP of psychotherapy sessions has progressed from proof-of-concept studies to automated pipelines and early feedback systems. The evidence suggests that selected therapist behaviors, particularly in MI, can be coded automatically with promising accuracy, and that automated feedback is acceptable to clinicians in initial studies. Evidence that such feedback improves therapeutic skills and, ultimately, client outcomes in routine care is still limited.

From a meaning-centered perspective, these tools deserve a differentiated assessment. They may help therapists perceive their own conversational habits more clearly and thus support reflective practice. They cannot, however, capture what a conversation means to the person who seeks help. Used as instruments of learning under conditions of informed consent and strict data protection, automated analyses may enrich training and quality assurance; used as substitutes for clinical judgment or as instruments of control, they risk reducing the therapeutic encounter to what can be counted.

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